Des rouleaux à la st Valentin? Étude du vol F10

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Scientific questions in my phd

- **Understanding physical processes** leading to the roll structures in the boundary layer. Which instabilities? (Etling and Brown 1993).

- Impact of structures in the **transport of momentum, moisture and heat** (Rivière, Ricard, and Arbogast 2020).

- Strong winds & **heat fluxes** : further study the relationship between the rolls' geometry and fluxes (Lfarh et al. 2024)

General presentation of the F10

Several crossings of a warm front and associated mid-level jet.

Flight of the 14/02/2026

Take off	14:03 UTC
Landing	18:10 UTC
S1 overpass	18:30 UTC
EarthCARE overpass	15:30 UTC

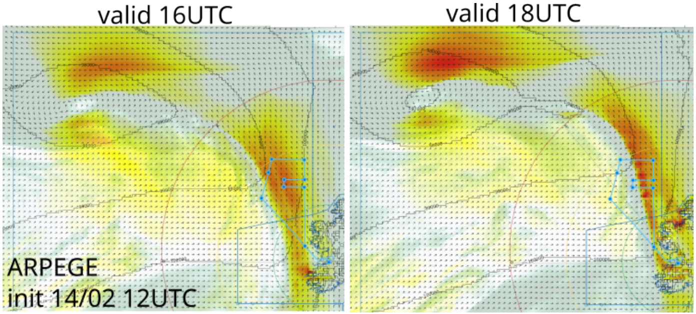

Figure 1: Wind@925hPa & mid-level cloud cover (AERIS)

- Organized structures on sea roughness signal
- Analyses to be done to retrieve and compare to LES simulation

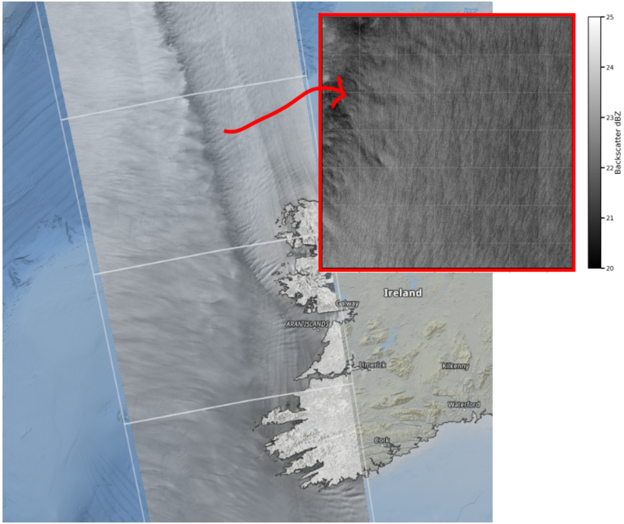

Figure 2: SAR data of S1-A at 18:03 UTC (image from ESA)

Patterns in the boundary layer on the RASTA reflectivity.

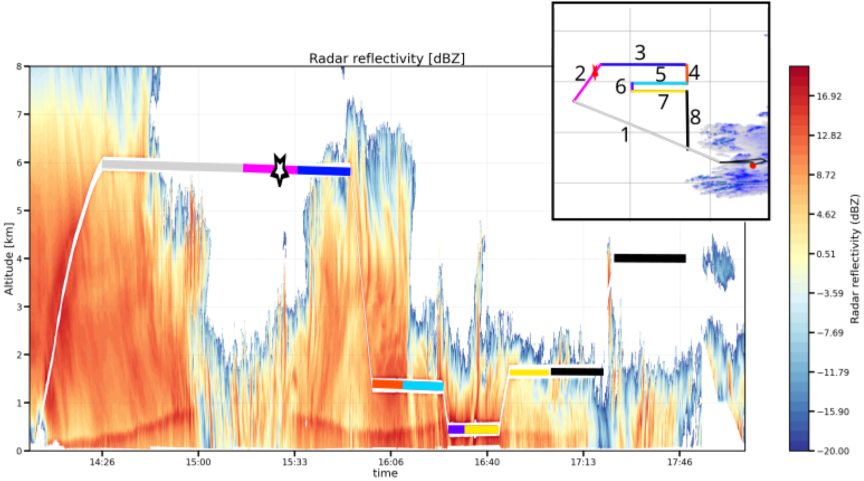

Figure 3: RASTA reflectivity (Julien Delanoë)

Vertical wind velocity can be retrieved from the RASTA signals.

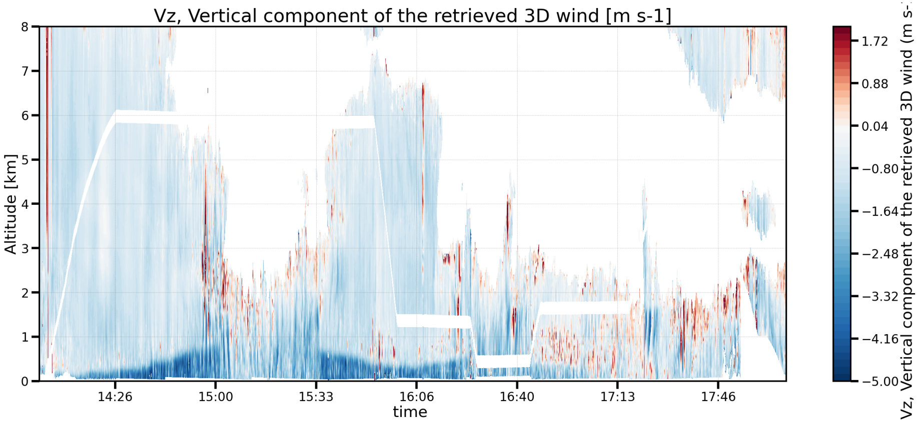

Figure 4: RASTA W-wind (Julien Delanoë)

Special interest in the lower legs : leg 5

Coherent structures seem to be detected by the RASTA (resolution 300m / 60m).

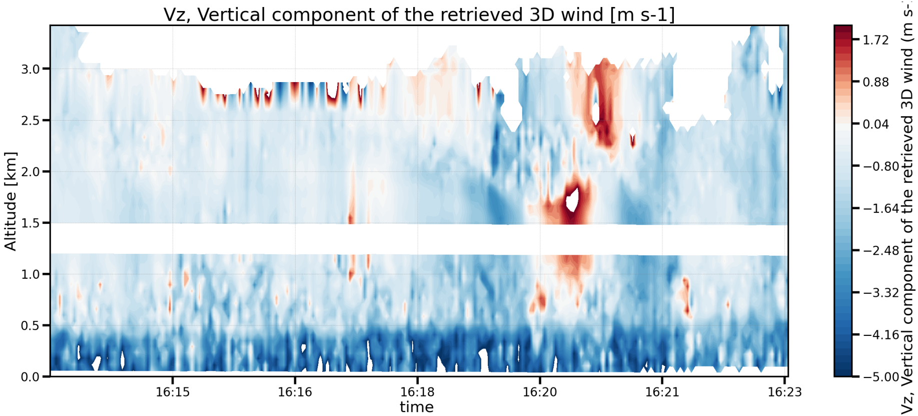

Figure 5: W-wind retrieved from RASTA signal in legs 5 (Julien Delanoë)

First Meso-NH realistic simulation at a resolution of 4 km.

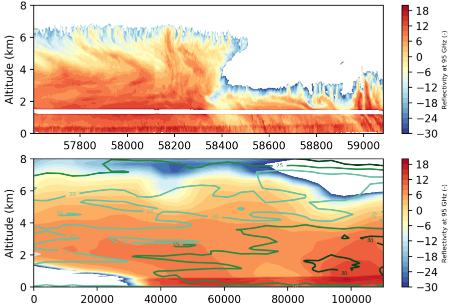

Figure 6: RASTA and simulation reflectivity comparison (JP Chaboureau)

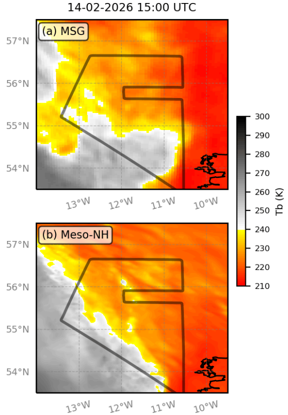

Figure 7: BT comparison between simulation and MSG (JP Chaboureau)

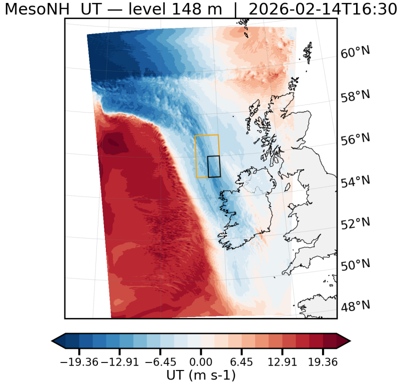

Figure 8: U-wind in the Meso-NH simulation

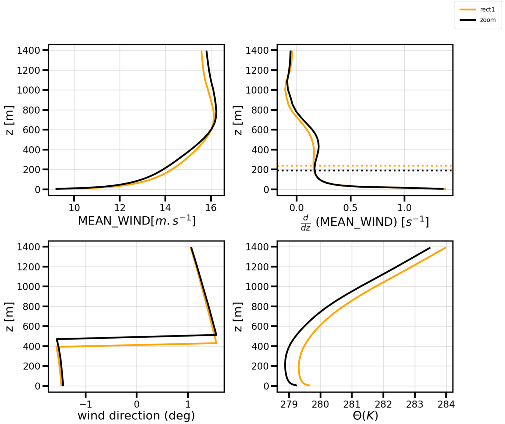

Figure 9: Mean profiles of wind and potential temperature

Objectives & next steps

The objectives of the study are :

- to **reproduce boundary layer roll structures** and compare Meso-NH simulations with observations in RASTA and SAR
- to study the relation between the SAR signal and the 3D wind structure
- to try to **quantify the impact of the structures in the transport** of momentum, moisture and heat.